

ENOSH REEDER

Mechatronics engineer with extensive CAD and prototyping experience across robotics, consumer electronics, and analytical test systems. Builds custom hardware, automation, and test tooling for device R&D.

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San Francisco, CA

TECHNICAL SKILLS

Design:	SolidWorks (CAD/CAM/FEA/PDM) Fusion 360 / Inventor / HSM Altium PCB design
Software:	Python C++ MATLAB SQL LabVIEW ROS2 / MoveIt ML / computer vision Linux Arduino G-code Star-CCM+ CFD
Fabrication:	CNC lathe / mill / router waterjet laser cutter FFF / FDM / SLA / MJF 3D printing carbon-fiber & fiberglass composites soldering MIG / SMA welding casting brazing cable harnesses
DFM/DFA:	Injection molding sheet metal CNC aluminum extrusion EDM
Certs/Lang:	ASME Y14.5 GD&T FAA Remote Pilot (sUAS) English (native) Mandarin Chinese (fluent)

ENGINEERING EXPERIENCE

Senior Mechatronics Engineer — JUUL | San Francisco, CA Aug 2024 – Current

- Developing robotic work cell to automate device handling (target ~4x throughput); Python and ROS2 / MoveIt control of UR10e arm with Intel RealSense depth camera and NVIDIA Jetson Orin Nano; prototyped CV on Hiwonder JetHexa hexapod
- Built custom hardware integrating device, flow sensor, and logger to capture synchronized user-session data
- Built an AI pipeline that transcribed and analyzed session audio (cloud speech + LLMs), correlating spoken feedback with flow and zone-temperature data to inform firmware tuning
- Retrofitted and integrated an FTIR spectrometer and a PI-TOF mass spectrometer for aerosol characterization (heated manifolds, inlets, solenoid puffing control, enclosures)
- Brought toxicology testing in-house by building a 12-way solenoid-manifold test rig, cutting per-test cost by six-figure margin
- Building in-house functional circuit tester (clamshell FCT fixture) for validating new device mainboards and a production-line QC tool that interfaces with an LCR meter for rapid impedance characterization of components
- Designed a magnetic pogo-pin charging/flushing dock and shipped ~150 units to external evaluation users (EUT)
- Altium PCBs: flexible RTD temperature-sensor board and rigid boards for flow measurement and USB-C integration
- Designed superelastic nitinol actuators with manual and automated proof-of-concept prototypes for a new R&D project; characterized custom valves, pumps, and microfluidic components

Metrology Engineer — QuantumScape | San Jose, CA Aug 2023 – Aug 2024

- Improved casting quality and yield on an R&D roll-to-roll slot-die coater for solid-state battery separators; designed and installed cleanroom optical linescan inspection systems and web cleaners
- Trained computer-vision models for automated defect detection and segmentation, reaching >75% recall and precision (independently validated against Keyence VK microscopy and in-house metrology)
- Evaluated X-ray and laser / 3D-microscope sensors and parsed inspection data in Python and SQL

Senior Mechatronics Engineer — Udelv | Burlingame, CA Sep 2021 – Jul 2023

- Designed, built, and tested the uPod, a 340 ft³ modular robotic cargo platform for autonomous electric delivery vehicles
- Engineered actuated cargo doors with thousands of moving parts, modular composite shelving, structures, and sensor housings; resolved failure modes for million-cycle reliability with motor current draw analysis and tribology material optimization
- Sourced and interfaced (DFM, DFA) with vendors for injection-molded parts, extrusions, and OTS components; used top-down SolidWorks modeling for complex parts and assemblies

Mechatronics Engineer — Transparent Sky | Albuquerque, NM Jul 2020 – Sep 2021

- Designed and built an aircraft-mounted, gigapixel-class Wide-Area Motion Imaging (WAMI) system in an aerodynamic composite pod for the Air Force Research Laboratory (AFRL) at ~98% lower cost than comparable government programs
- Evaluated sensors and lenses for day/night flight testing; streamlined in-flight test setup (~90% faster)
- Hired and mentored junior engineers and the fabrication team; built out the entire composite shop from scratch (4×8 ft CNC router for mold-making, autoclave, curing ovens, paint/sanding booth)

NSF Research Intern — UC San Diego — Bio-Inspired Robotics & Design Lab | La Jolla, CA Jun 2019 – Aug 2019

- Designed everting soft-robotics mechanisms for computer vision on an ROV; generated point clouds via photogrammetry and LiDAR to track mangrove populations over time.

EDUCATION & LEADERSHIP

California Polytechnic State University (Cal Poly) | San Luis Obispo, CA 2017 – 2020

B.S. Aerospace Engineering — Aeronautics Concentration

FIRST Robotics Team 171 President | Platteville, WI 2013 – 2017